Mechanical Engineering Case Study

- Beam Deflection

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Step 1: Problem Identification and Statement

This assignment aims to develop a program using MATLAB to calculate a beam's vertical deflection at a position under a certain amount of load. And plot the graph of the result of the beam deflection.

Step 2: Gathering Information and input/Output diagram

- *Gathering of information*

Deflection in an engineering definition refers to the movement of a beam from its original position due to the loads applied to the member. This displacement can occur when an external load is applied. The deflection of a beam is calculated based on a variety of factors, therefore it can be simplified into simple deflection formula for quick calculations.
In this assignment, the beam will be supported at both ends, and a load (P) will be applied at a point on its length (L). Due to this load, the beam will deflect vertically, this vertical deflection (V) of the beam will vary along the length of the beam from x=0 to x=L.
Two simple formulas will be used to calculate the deflection of the beam.
If the x value is within the range of the start of the beam to where the load is applied (a), the formula will be V=((P*b)/(6*E*I*L))*(((-L^2)+(b^2))*x+(x^3)).
If the x value is from where the load is applied (a) to the end of the beam, the formula will be V=((P*b)/(6*E*I*L))*(((-L^2)+(b^2))*x+(x^3)-(L/b)*((x-a)^3)).
Definition of the symbols used in the formulas:
x: the distance from the left end
L: length of the beam
P: the load
a: the location where the load is applied
E: Young's modulus of the beam material
I: the second moment of the area

The value of the load could be negative since its sign indicates the direction the load is applied (from above the beam as positive and below the beam as negative). The rest values should be positive. Variable x and a should be within the length of the beam (0<x&a<L).

- *Input/Output Description*

Input:

The inputs are the selection of the number in the menu displayed on the screen

1. Input '1' if the user wants to calculate the beam deflection at position x and plot its corresponding graph.
2. Input '2' if the user wants to plot the graph only

Both selections lead to the following inputs.

1. Distance from the left end
2. Length of the beam
3. The load
4. The location where the load is applied
5. Young's modulus of the beam material
6. The second moment of the area

Invalid inputs include: the distance from the left end and the location where the load is applied are outside the length of the beam, and values other than the load are negative.

Output:

1. A menu is displayed as output with two options
2. Corresponding to input 1

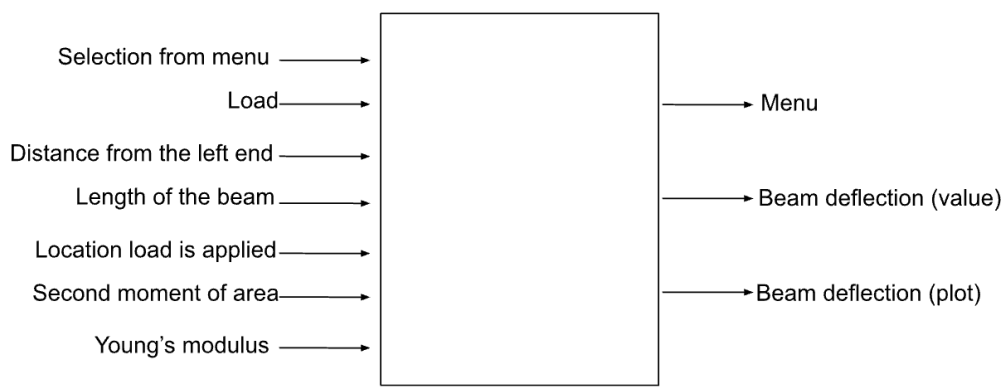
Outputs are the calculated beam deflection value and the graph.

3. Corresponding to input 2

Output is the plotting of the resulting beam deflection graph.

4. Error message if the user's menu selection or input is invalid

Inputs and outputs are also shown in the diagram below:



Step 3: Test cases and Algorithm Design

- *Test Cases*

Sample of the main menu

Please make your selection: 1) Calculation of the beam deflection and graph plotting 2) Plot the graph of beam deflection

Test Case 1: Invalid input in the menu selection

In the menu, users need to select the function they wish the program to do. If the user enters an invalid input (numbers other than 1, or 2), the program will display an error message and prompt the user for another valid value.

Test Case 2: Selecting 1) Calculation of the beam deflection and graph plotting
x is between the start of the beam to where the load is applied

If the user selects 1) Calculation of the beam deflection and graph plotting, the program will prompt the user for values of x, L, a, P, E, and I. The values tested here are: x=2.5; L=5; a=3; E=30*10^6; I=0.0256; P=30000.
Since the value x is smaller than 'a', it means x is between the start of the beam to where the load is applied, the first formula will be used to calculate the beam deflection. If the load is positive (applying from up downwards), the beam deflection should be negative showing it is pushed down by the load.
An x-V graph will be displayed on the screen with the y-axis being the beam deflection and the x-axis being the length of the beam, all values calculated based on the user's inputs. Proper title and x-, and y-axis labels should be displayed on the graph.
A goodbye message will be displayed.

Test Case 3: Selecting 1) Calculation of the beam deflection and graph plotting
x is between where the load is applied to the end of the beam

Same as test case 2, continue as the following.
The values going to be tested here are: x=4.05; L=5; a=3; E=30*10^6; I=0.0256; P=30000
Since the value x is larger than 'a', it means x is between where the load is applied to the end of the beam, the second formula will be used to calculate the beam deflection. If the load is positive (applying from up downwards), the beam deflection should be negative showing it is pushed down by the load.
Graph plotting same as test case 2.
A goodbye message will be displayed.

Test Case 4: Selecting 2) Plot the graph of beam deflection

If the user selects 2) Plot the graph of beam deflection. Same as test case 2, values in test case 2 will be used. No outputs of beam deflection values. Only the x-V graph will be displayed. The graph will have its y-axis being the beam deflection and the x-axis being the length of the beam, all values calculated based on the user's inputs. Proper title and x-, and y-axis labels should be displayed on the graph.
A goodbye message will be displayed.

Test Case 5: Invalid inputs in variables

This test case will test all the following invalid inputs. Everything other than the load is negative. The distance from the left end (x), or the location where the load is applied (a) is outside the length of the beam. All these invalid inputs will result in an error message, and the program will prompt the user for another value.

- *Algorithm Design*

```
Print "Welcome to the program, this program performs: ", newline
Print "1) Calculation of beam deflection at x and graph plotting", newline
Print "2) Plot the graph of beam deflection", newline
Print "Please make your selection: ", newline
Read value into ctr
While ctr>2 OR ctr<1, repeat the following
        Print "Your input is invalid, please make your selection: ", newline
        Read value into ctr

Print "Please enter the length of the beam: ", newline
Read value into L
While L<0, repeat the following
        Print "Your input is invalid, please enter a valid length of the beam: ", newline
        Read value into L
Print "Please enter the distance from the left end: ", newline
Read value into x
Assign 0 to L with a step of 0.01 to validx
While x is not equal to any value within validx, repeat the following
        Print "Your input is invalid, please enter a valid distance from the left end: ", newline
        Read value into x
Print "Please enter the load: ", newline
Read value into P
Print "Please enter the location where the load is applied: ", newline
Read value into a
```

```
Assign 0 to L with a step of 0.01 to valida
While a is not equal to any value within valida, repeat the following
        Print "Your input is invalid, please enter a valid location of where the load is applied: ",
                newline
        Read value into a
Print "Please enter the Youngs modulus of the beam material: ", newline
Read value into E
While E<0, repeat the following
        Print "Your input is invalid, please enter a valid Youngs modulus of the beam material: ",
                newline
        Read value into E
Print "Please enter the second moment of area: ", newline
Read value into I
While I<0, repeat the following
        Print "Your input is invalid, please enter a valid second moment of area: ", newline
        Read value into I
Assign L-a to b

if ctr=1
if (0<x AND x<a)
   Assign ((P*b)/(6*E*I*L))*(((-L^2)+(b^2))*x+(x^3)) to V
   Print "The calculated deflection at x= (keep space for a float variable with one decimal
           place) is (keep space for a float variable with six decimal places)", newline
           Assign x to the first spot and V to the second spot
Otherwise if (a<x AND x<L)
                Assign ((P*b)/(6*E*I*L))*(((-L^2)+(b^2))*x+(x^3)-(L/b)*((x-a)^3)) to V
                Print "The calculated deflection at x= (keep space for a float variable with one
                        decimal place) is (keep space for a float variable with six decimal places)",
                        newline
                        Assign x to the first spot and V to the second spot
Assign 0 to a with a step of 0.05 to X
Assign ((P.*b)/(6.*E.*I.*L)).*(((-L.^2)+(b.^2)).*X+(X.^3)) to V
Plot the graph with X as the x-axis, V as the y-axis, green as the color with a solid line, and
values marked by asterisks
Keep the graph
Title of the graph is "Graph of the Beam Deflection"
X-axis label of the graph is "Length of the beam"
Y-axis label of the graph is "Beam deflection"
Assign a to L with a step of 0.05 to X
Assign ((P.*b)/(6.*E.*I.*L)).*(((-L.^2)+(b.^2)).*X+(X.^3)-(L./b).*((X-a).^3)) to V
```

Plot the graph with X as the x-axis, V as the y-axis, green as the color with a solid line, and values marked by asterisks
Release the graph
Print “Thank you for using the program, goodbye!”, newline

if ctr=2
Assign 0 to a with a step of 0.05 to X
Assign ((P.*b)/(6.*E.*I.*L)).*(((-L.^2)+(b.^2)).*X+(X.^3)) to V
Plot the graph with X as the x-axis, V as the y-axis, green as the color with a solid line, and values marked by asterisks
Keep the graph
Title of the graph is “Graph of the Beam Deflection”
X-axis label of the graph is “Length of the beam”
Y-axis label of the graph is “Beam deflection”
Assign a to L with a step of 0.05 to X
Assign ((P.*b)/(6.*E.*I.*L)).*(((-L.^2)+(b.^2)).*X+(X.^3)-(L./b).*((X-a).^3)) to V
Plot the graph with X as the x-axis, V as the y-axis, green as the color with a solid line, and values marked by asterisks
Release the graph
Print “The graph has displayed, thank you for using the program, goodbye!”, newline

Step 4: Code of implementation

```
%This program aims to calculate the beam deflection with multiple provided
%values, and plot its corresponding x-V graph.
fprintf('Welcome to the program, this program performs: \n');
fprintf('1) Calculation of beam deflection at x and graph plotting \n');
fprintf('2) Plot the graph of beam deflection \n');
ctr=input('Please make your selection: \n');
%check the validity of menu selection
while ctr>2||ctr<1
    ctr=input('Your input is invalid, please make your selection: \n');
end

L=input('Please enter the length of the beam: \n');
%the length of the beam cannot be negative
while L<0
    L=input('Your input is invalid, please enter a valid length of the beam:
\n');
end
x=input('Please enter the distance from the left end: \n');
%x is only valid if it is between 0 and L
validx=0:0.01:L;
%if the x input is not equal to any number between 0 and L with steps of
%0.01, it is invalid
while ~any(x==validx)
```

```
    x=input('Your input is invalid, please enter a valid distance from the
left end: \n');
end
%P can be negative which in this case indicates applying the load from down
upwards to
%the beam
P=input('Please enter the load: \n');
a=input('Please enter the location where the load is applied: \n');
%a is only valid if it is between 0 and L
valida=0:0.01:L;
%if the a input is not equal to any number between 0 and L with steps of
%0.01, it is invalid
while ~any(a==valida)
    a=input('Your input is invalid, please enter a valid location of where
the load is applied: \n');
end
E=input('Please enter the Youngs modulus of the beam material: \n');
%Youngs modulus cannot be negative
while E<0
    E=input('Your input is invalid, please enter a valid Youngs modulus of
the beam material: \n');
end
I=input('Please enter the second moment of area: \n');
%the second moment of area cannot be negative
while I<0
    I=input('Your input is invalid, please enter a valid second moment of
area: \n');
end
b=L-a;

%if the user wants to perform both calculation and graph plotting
if(ctr==1)
%x is between 0 and a
if(0<x&&x<a)
    V=((P*b)/(6*E*I*L))*(((-L^2)+(b^2))*x+(x^3));
    fprintf('The calculated deflection at x=%.1f is %.6f\n', x, V);
%x is between a and L
elseif(a<x&&x<L)
    V=((P*b)/(6*E*I*L))*(((-L^2)+(b^2))*x+(x^3)-(L/b)*((x-a)^3));
    fprintf('The calculated deflection at x=%.1f is %.6f\n', x, V);
end
X=(0:0.05:a);
%calculating V for all x values between 0 and a with a step of 0.05
V=((P.*b)/(6.*E.*I.*L)).*(((-L.^2)+(b.^2)).*X+(X.^3));
plot(X,V, 'g-*')
%keep this scene and the following graph can be plotted in the same scene,
%thus producing a continuous graph
hold on
%formatting the graph
title('Graph of the Beam Deflection')
xlabel('Length of the beam')
ylabel('Beam deflection')
X=(a:0.05:L);
%calculating V for all x values between a and L with a step of 0.05
V=((P.*b)/(6.*E.*I.*L)).*(((-L.^2)+(b.^2)).*X+(X.^3)-(L./b).*((X-a).^3));
```

```
plot(X,V, 'g-*')
hold off
fprintf('Thank you for using the program, goodbye!');
end

%if the user only wants to perform graph plotting
if(ctr==2)
X=(0:0.05:a);
%calculating V for all x values between 0 and a with a step of 0.05
V=((P.*b)/(6.*E.*I.*L)).*(((-L.^2)+(b.^2)).*X+(X.^3));
plot(X,V, 'g-*')
%keep this scene and the following graph can be plotted in the same scene,
%thus producing a continuous graph
hold on
%formatting the graph
title('Graph of the Beam Deflection')
xlabel('Length of the beam')
ylabel('Beam deflection')
X=(a:0.05:L);
%calculating V for all x values between a and L with a step of 0.05
V=((P.*b)/(6.*E.*I.*L)).*(((-L.^2)+(b.^2)).*X+(X.^3)-(L./b).*((X-a).^3));
plot(X,V, 'g-*')
hold off
fprintf('The graph has displayed, thank you for using the program,
goodbye!');
end
```

Step 5: Test and Verification

Menu display

```
Command Window
>> assignment4
Welcome to the program, this program performs:
1) Calculation of beam deflection at x and graph plotting
2) Plot the graph of beam deflection
Please make your selection:
```

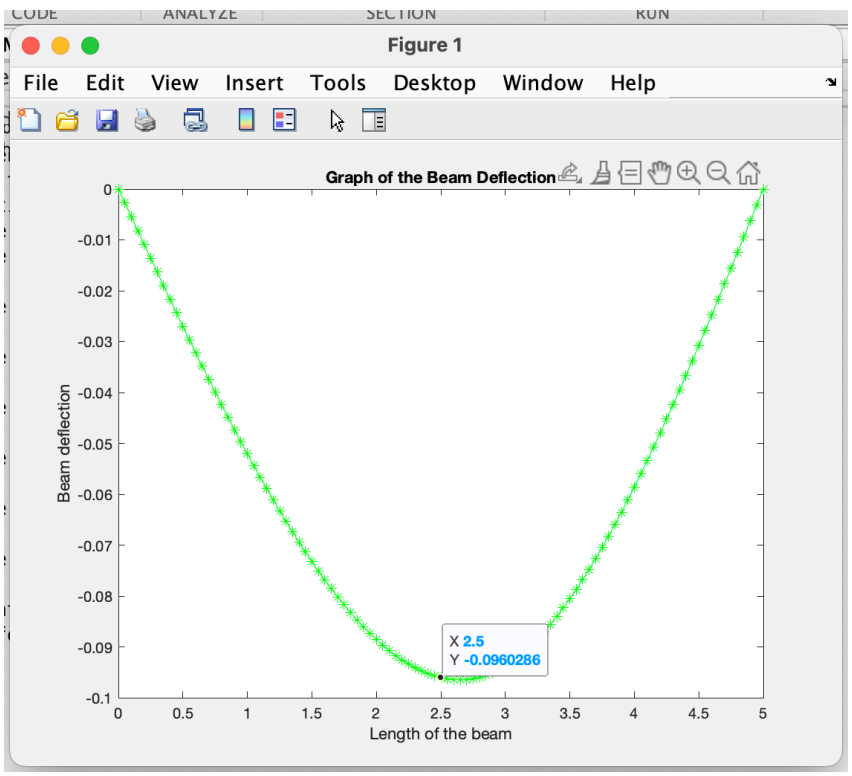
Test Case 1: Invalid input in the menu selection

```
Command Window
>> assignment4
Welcome to the program, this program performs:
1) Calculation of beam deflection at x and graph plotting
2) Plot the graph of beam deflection
Please make your selection:
0
Your input is invalid, please make your selection:
4
Your input is invalid, please make your selection:
```

(The user's inputs are 0 and 4, which are invalid, an error message is displayed, and asked the user for another valid value.)

Test Case 2: Selecting 1) Calculation of the beam deflection and graph plotting x is between the start of the beam to where the load is applied

```
Command Window
>> assignment4
Welcome to the program, this program performs:
1) Calculation of beam deflection at x and graph plotting
2) Plot the graph of beam deflection
Please make your selection:
1
Please enter the length of the beam:
5
Please enter the distance from the left end:
2.5
Please enter the load:
30000
Please enter the location where the load is applied:
3
Please enter the Youngs modulus of the beam material:
30*10^6
Please enter the second moment of area:
0.0256
The calculated deflection at x=2.5 is -0.096029
Thank you for using the program, goodbye!>>
```

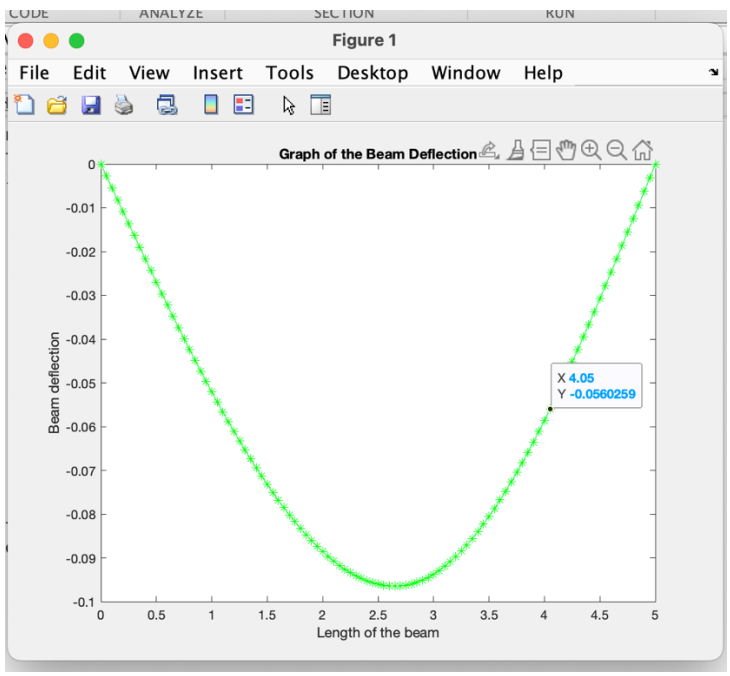


(The user's inputs are: x=2.5; L=5; a=3; E=30*10^6; I=0.0256; P=30000.
Since the value x is larger than 0 and smaller than 'a', the first formula was used to calculate the beam deflection. Successfully displayed to the user. A corresponding graph is successfully displayed with proper title, x-, and y-axis labels. Goodbye message displayed.)

Notably, the graph is an upward parabola with a negative value of the beam deflection, this is due to a positive load pushing the beam down.

Test Case 3: Selecting 1) Calculation of the beam deflection and graph plotting x is between where the load is applied to the end of the beam

```
Command Window
>> assignment4
Welcome to the program, this program performs:
1) Calculation of beam deflection at x and graph plotting
2) Plot the graph of beam deflection
Please make your selection:
1
Please enter the length of the beam:
5
Please enter the distance from the left end:
4.05
Please enter the load:
30000
Please enter the location where the load is applied:
3
Please enter the Youngs modulus of the beam material:
30*10^6
Please enter the second moment of area:
0.0256
The calculated deflection at x=4.0 is -0.056026
Thank you for using the program, goodbye!>>
```

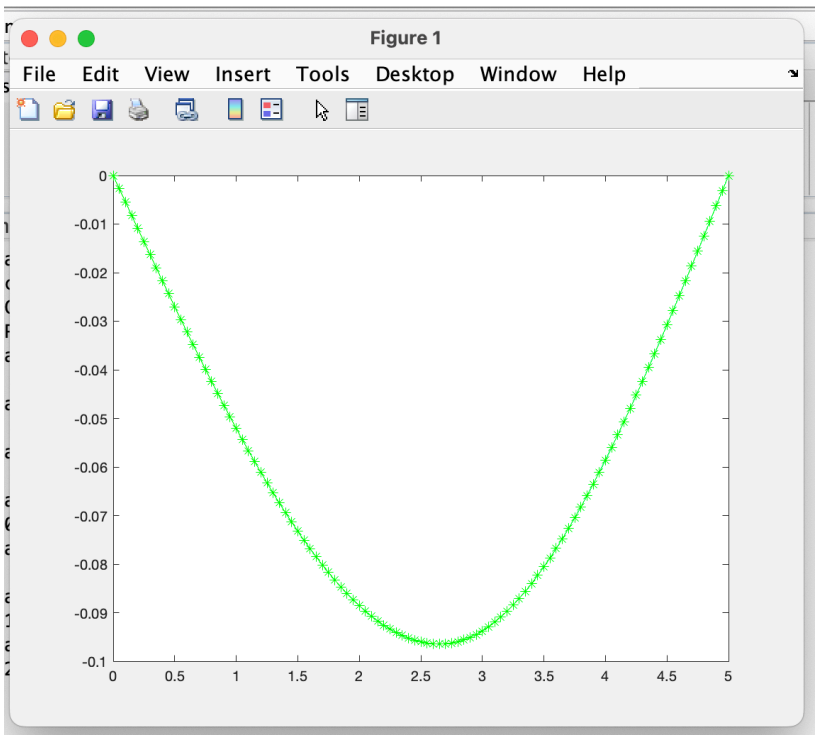


(The user's inputs are: x=4.05; L=5; a=3; E=30*10^6; I=0.0256; P=30000.
Since the value x is larger than 'a' and smaller than L, the second formula was used to calculate the beam deflection. Successfully displayed to the user. A corresponding graph is successfully displayed with proper title, x-, and y-axis labels. Goodbye message displayed.)

Notably, the graph is an upward parabola with a negative value of the beam deflection, this is due to a positive load pushing the beam down.

Test Case 4: Selecting 2) Plot the graph of beam deflection

```
Command Window
>> assignment4
Welcome to the program, this program performs:
1) Calculation of beam deflection at x and graph plotting
2) Plot the graph of beam deflection
Please make your selection:
2
Please enter the length of the beam:
5
Please enter the distance from the left end:
2.5
Please enter the load:
30000
Please enter the location where the load is applied:
3
Please enter the Youngs modulus of the beam material:
30*10^6
Please enter the second moment of area:
0.0256
The graph has displayed, thank you for using the program, goodbye!>>
```



(The user's inputs are: x=2.5; L=5; a=3; E=30*10^6; I=0.0256; P=30000.
A corresponding graph is successfully displayed with proper title, x-, and y-axis labels. Goodbye message displayed.)

Notably, the graph is an upward parabola with a negative value of the beam deflection, this is due to a positive load pushing the beam down.

Test Case 5: Invalid inputs in variables

```
Command Window
>> assignment4
Welcome to the program, this program performs:
1) Calculation of beam deflection at x and graph plotting
2) Plot the graph of beam deflection
Please make your selection:
1
Please enter the length of the beam:
-1
Your input is invalid, please enter a valid length of the beam:
5
Please enter the distance from the left end:
6
Your input is invalid, please enter a valid distance from the left end:
2.5
Please enter the load:
30000
Please enter the location where the load is applied:
6
Your input is invalid, please enter a valid location of where the load is applied:
3
Please enter the Youngs modulus of the beam material:
-10
Your input is invalid, please enter a valid Youngs modulus of the beam material:
30*10^6
Please enter the second moment of area:
-3
Your input is invalid, please enter a valid second moment of area:
0.0256
The calculated deflection at x=2.5 is -0.096029
Thank you for using the program, goodbye!>>
```

(The user's inputs include all the invalid values: negative for everything other than the load; the distance from the left end (x), and the location where the load is applied (a) are outside the length of the beam. The error message displayed successfully and prompted the user for another value.)